

The Role of Orthogonality in Additive GPs for Interpretability and Predictive Power

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The Curse of Dimensionality and Additive Models

- ▶ **Curse of Dimensionality:** In high-dimensional spaces, the number of points required grows exponentially with the number of dimensions. e.g. consider number of grid point in N dimensions
- ▶ **Implication for ML:** Machine learning models succeed, therefore they require only a subset of the input space, not an exhaustive exploration of all points.
- ▶ **Additive GPs:** Allow us to directly decompose the potential function space into specific lower-dimensional components.

Additive Models to Additive Gaussian Processes

Step 1: Additive Models (AMs)

- ▶ Additive Models represent the response Y as a sum of functions of each individual predictor:

$$Y = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_D(X_D) + \epsilon$$

where ϵ is the noise term.

- ▶ The benefit of AMs is interpretability: the effect of each predictor can be understood individually.
- ▶ Typically continuous.

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Step 2: Generalized Additive Models (GAMs)

- ▶ GAMs extend AMs by introducing a link function g , which allows for a wider range of response types:

$$g(\mathbb{E}[Y]) = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_D(X_D)$$

- ▶ GAMs provide the flexibility to model non-normal response distributions, such as binary outcomes or count data.

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Step 3: Additive Gaussian Processes (GPs)

- ▶ Additive GPs take the concepts of AMs and GAMs into a probabilistic framework by considering each function $f_i(X_i)$ to be a random function drawn from a Gaussian Process:

▶

$$f(\mathbf{x}) = f_0 + \sum_{i=1}^D f_i(x_i) + \sum_{i < j} f_{ij}(x_i, x_j) + \cdots + f_{1,2 \dots D}(x_1 \dots x_D)$$

- ▶ GPs quantify uncertainty in predictions, which is not directly addressed in AMs or GAMs.
- ▶ So we keep the interpretability, but extend this to interaction terms

Background Context

- ▶ **Paper 1: Additive Gaussian Processes** (*Duvenaud et al.*)
 - ▶ This paper unifies GAMs and SE-GPs using additive kernels.

- ▶ **Paper 2: Additive Gaussian Processes Revisited** (*Lu et al.*)
 - ▶ This paper introduces orthogonality constraints for unique decomposition.

Identifiability Challenges in Additive Models

- ▶ Consider a two-dimensional function decomposed as:
 $f(x_1, x_2) = f_1(x_1) + f_2(x_2)$
- ▶ An alternative decomposition is:
 $f(x_1, x_2) = (f_1(x_1) + \Delta) + (f_2(x_2) - \Delta), \quad \forall \Delta$

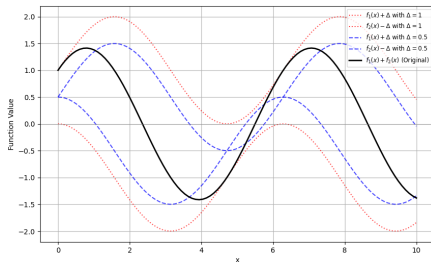


Figure 1: Illustration of infinite decompositions.

Orthogonality Constraints

Zero Mean and Covariance Constraints

- ▶ **Zero Mean Constraint:**

$$\int_{x_i} f_i(x_i) p_i(x_i) dx_i = 0, \quad \forall i \in [D]$$

- ▶ **Zero Covariance Constraint:**

$$\text{Cov}[f_u(x_u), f_v(x_v)] = 0, \quad \text{for } u \neq v$$

- ▶ These constraints help maintain interpretability by uniquely defining the contribution of each component.

Problem Statement

Core Question

Does enforcing uniqueness in Gaussian Process components lead to better predictive performance?

- ▶ Yes! We observe improvement with orthogonality, but the reason behind this improvement is still unclear.
- ▶ Why should we try to understand this?

Observed Benefits of Orthogonality

Empirical Results

- ▶ Improved Predictive Accuracy: Models with orthogonality constraints show improved performance, such as reduced Mean Squared Error (MSE).
- ▶ Hierarchy of Main and Interaction Effects: Enforcing orthogonality has also been shown to reduce the number of terms needed.

Reported Results 1 - RMSE

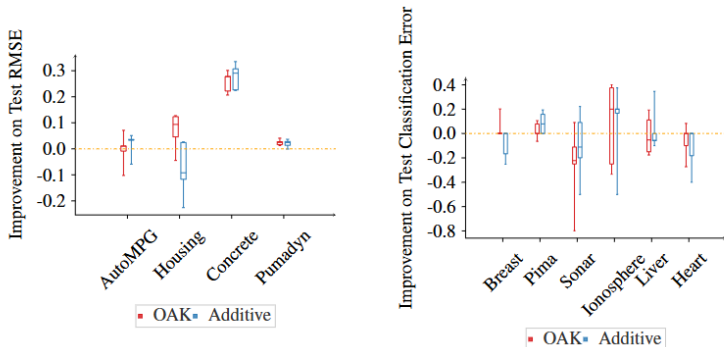


Figure 2: Relative RMSE improvement for regression (left) and classification (right) across datasets for OAK vs Additive Kernels. Taken from Lu et al. (2022).

Reported Results 2 - Hierarchy Reduction

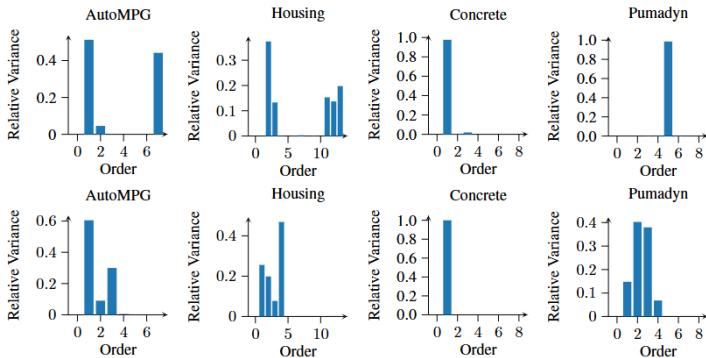


Figure 3: Top row uses Additive Kernel from Duvenaud et al. (2011); Bottom row uses OAK (orthogonal additive kernel) from Lu et al.(2022). At similar performance levels, OAK requires lower dimensional order interactions. Figure from Lu et al.(2022).

Project Outline

Completed tasks

- ▶ Literature Review
- ▶ Considered and proved subsets of regressions tasks in which the constraint holds

How we proceed

- ▶ Finish verifying Additive GP models with and without standard orthogonality constraints across datasets.
- ▶ Investigate mechanism for the improvement.

Summary

Key Takeaways

- ▶ Enforcing uniqueness and orthogonality enhances the interpretability of Gaussian Process models.
- ▶ Current constraints show improved performance, but understanding why this is the case remains a key research goal.

Thank you!

Questions?

Hierarchy Decomposition

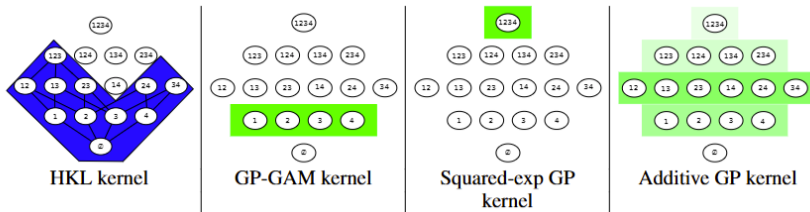


Figure 4: Figure from Duvenaud et al. (2011)

Newton-Girard Formulation

The Newton-Girard recursive relations are given by:

$$k_{add_n} = e_n(z_1, \dots, z_D) = \frac{1}{N} \sum_{k=1}^n (-1)^{k-1} e_{n-k}(z_1, \dots, z_D) s_k(z_1, \dots, z_D)$$

where:

- ▶ $e_0 = 1$
- ▶ $s_k(z_1, \dots, z_D) = \sum_{i=1}^D z_i^k$
- ▶ $e_{n-k}(z_1, \dots, z_D)$ represents the elementary symmetric polynomial of degree $n - k$